

# DR. PRATIK GAJANE

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## PROFESSIONAL SUMMARY

- **Accomplished machine learning researcher** with a PhD from the University of Lille and extensive expertise in adaptive sequential decision-making and responsible machine learning for human-centric applications.
- **Extensive publication record** at top conferences such as ICML, IJCAI, COLT, ICDM, and AAMAS.
- **Experienced educator** with a proven track record in designing and teaching undergraduate and postgraduate modules (receiving excellent teaching evaluations), as well as developing curricula. Holding a University Teaching Qualification (UTQ/BKO) from the Netherlands.
- **Supervisor of PhD and MSc students**, with a history of successful supervision leading to international publications.
- **Experience in designing and delivering impactful interdisciplinary research** through world-wide collaborations with universities in the Netherlands, France, Germany, Austria, Belgium, the USA, and India, as well as industry leaders such as KPN, ASML, and Orange Labs.
- **Experience in leading collaborative research projects and engaging with cross-disciplinary stakeholders.**
- **Active contributor to university initiatives and administrative tasks.**

## ACADEMIC EXPERIENCE

|                     |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| <b>2024 to -</b>    | <b>University of Orleans (France)</b>                                        |
| <b>POSITION</b>     | <b>Chair Junior Professor (with a research funding of €200,000 from ANR)</b> |
| <b>2021 to 2024</b> | <b>Eindhoven University of Technology (The Netherlands)</b>                  |
| <b>POSITION</b>     | <b>Postdoctoral researcher</b>                                               |
| <b>2018 to 2021</b> | <b>Montanuniversität Leoben (Austria)</b>                                    |
| <b>POSITION</b>     | <b>Postdoctoral researcher</b>                                               |

## EDUCATION

|                      |                                                                             |
|----------------------|-----------------------------------------------------------------------------|
| <b>2014 to 2017</b>  | <b>INRIA Lille-team SequeL, Université Lille &amp; Orange labs (France)</b> |
| <b>QUALIFICATION</b> | <b>PhD</b>                                                                  |
| <b>DEFENSE DATE</b>  | Nov 2017                                                                    |
| <b>TOPIC</b>         | Sequential Learning with Partial Feedback <a href="#">↗</a>                 |
| <b>2012 to 2014</b>  | <b>Indian Institute of Technology Madras (India)</b>                        |
| <b>QUALIFICATION</b> | <b>Master of Technology in Computer Science, CGPA : 9.19/10</b>             |
| <b>THESIS</b>        | Methods for the Multi-Armed Bandit problem                                  |
| <b>2005 to 2009</b>  | <b>University of Pune (India)</b>                                           |
| <b>QUALIFICATION</b> | <b>Bachelor of Engineering in Computer Science, First Class</b>             |

## PROFESSIONAL QUALIFICATIONS

|                         |                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Obtained in 2023</b> | <b>University Teaching Qualification (UTQ/BKO) <a href="#">↗</a></b><br>(Nationally accredited teaching certificate in the Netherlands) |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

## PREPRINTS

- [1] Pratik Gajane, Sean Newman, Mykola Pechenizkiy and John D. Piette. Investigating Gender Fairness in Machine Learning-Driven Personalized Care for Chronic Pain, [↗](#).
- [2] Danil Provodin, Pratik Gajane, Mykola Pechenizkiy and Maurits Kaptein. Provably Efficient Exploration in Constrained Reinforcement Learning: Posterior Sampling Is All You Need, [↗](#).
- [3] Pratik Gajane, Akshat Saxena, Maryam Tavakol, George Fletcher and Mykola Pechenizkiy. Survey on Fair Reinforcement Learning: Theory and Practice, [↗](#).
- [4] Pratik Gajane, Ronald Ortner, Peter Auer and Csaba Szepesvari. Autonomous Exploration for Navigating in Non-stationary Controlled Markov Processes, [↗](#).

- [5] Pratik Gajane. Adversarial Multi-dueling Bandits. In *the Workshop on Models of Human Feedback for AI Alignment at ICML*, 2024 [↗](#).
- [6] Ronald C. van den Broek, Rik Litjens, Tobias Sagis, Luc Siecker, Nina Verbeeke and Pratik Gajane. Multi-Armed Bandits with Generalized Temporally-Partitioned Rewards. In *the proceedings of the 22nd Symposium on Intelligent Data Analysis (IDA)*, 2024 [↗](#).
- [7] Jiong Li and Pratik Gajane. Curiosity-driven Exploration in Sparse-reward Multi-agent Reinforcement Learning. In *the 16th European Workshop on Reinforcement Learning (EWRL)*, 2023 [↗](#).
- [8] Ronald C. van den Broek, Rik Litjens, Tobias Sagis, Luc Siecker, Nina Verbeeke and Pratik Gajane. Multi-Armed Bandits with Generalized Temporally-Partitioned Rewards. In *the 16th European Workshop on Reinforcement Learning (EWRL)*, 2023 [↗](#).
- [9] Rosa van Tuijn, Tianqin Lu, Emma Driesse, Koen Franken, Pratik Gajane and Emilia Barakova. WeHeart: A Personalized Recommendation Device for Physical Activity Encouragement and Preventing “Cold Start” in Cardiac Rehabilitation. In *the proceedings of the 19th International Conference on Human-Computer Interaction (INTERACT)*, 2023 [↗](#).
- [10] Pratik Gajane, Peter Auer and Ronald Ortner. Autonomous Exploration for Navigating in MDPs using Blackbox RL Algorithms. In *the proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI)*, 2023 [↗](#).
- [11] Dennis Collaris, Pratik Gajane, Joost Jorritsma, Jarke J. van Wijk and Mykola Pechenizkiy. LEMON: Alternative Sampling for More Faithful Explanation through Local Surrogate Models. In *the proceedings of the 21st Symposium on Intelligent Data Analysis (IDA)*, 2023 [↗](#) ([Runner-up Frontier Prize](#) [↗](#)).
- [12] Rosa van Tuijn, Tianqin Lu, Emma Driesse, Koen Franken, Pratik Gajane and Emilia Barakova. WeHeart: A Personalized Recommendation Device for Physical Activity Encouragement and Preventing “Cold Start” in Cardiac Rehabilitation (extended abstract). In *the 2nd International Conference on Hybrid Human-Artificial Intelligence (HHAI)*, 2023 [↗](#).
- [13] Pratik Gajane. Local Differential Privacy for Sequential Decision Making in a Changing Environment. In *AAAI Privacy Preserving Artificial Intelligence (PPAI)*, 2023 [↗](#).
- [14] Danil Provodin, Pratik Gajane, Mykola Pechenizkiy and Maurits Kaptein. An Empirical Evaluation of Posterior Sampling for Constrained Reinforcement Learning. In *the Reinforcement Learning for Real Life Workshop at NeurIPS*, 2022 [↗](#).
- [15] Danil Provodin, Pratik Gajane, Mykola Pechenizkiy and Maurits Kaptein. The Impact of Batch Learning in Stochastic Linear Bandits. In *the proceedings of the 22nd International Conference on Data Mining (ICDM)*, 2022 [↗](#).
- [16] Danil Provodin, Pratik Gajane, Mykola Pechenizkiy and Maurits Kaptein. The Impact of Batch Learning in Stochastic Bandits. In *the workshop on Ecological Theory of Reinforcement Learning at NeurIPS 2021* [↗](#).
- [17] Filipo Studzinski Perotto, Sattar Vakili, Pratik Gajane, Yaser Faghan and Mathieu Bourgaïs. Gambler Bandits and the Regret of Being Ruined. In *the proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021 [↗](#).
- [18] Ronald Ortner, Pratik Gajane and Peter Auer. Variational Regret Bounds for Reinforcement Learning. In *the proceedings of the 35th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2020 [↗](#).
- [19] Peter Auer, Pratik Gajane and Ronald Ortner. Adaptively Tracking the Best Bandit Arm with an Unknown Number of Distribution Changes. In *the proceedings of the 32nd Annual Conference on Learning Theory (COLT)*, 2019 [↗](#).
- [20] Peter Auer, Yifang Chen, Pratik Gajane, Chung-Wei Lee, Haipeng Luo, Ronald Ortner and Chen-Yu Wei. Achieving Optimal Dynamic Regret for Non-stationary Bandits without Prior Information. In *the proceedings of the 32nd Annual Conference on Learning Theory (COLT)*, 2019 [↗](#).
- [21] Pratik Gajane, Ronald Ortner and Peter Auer. A Sliding-Window Approach for Reinforcement Learning in MDPs with Arbitrarily Changing Rewards and Transitions. In *Lifelong Learning: A Reinforcement Learning Approach Workshop at FAIM 2018* [↗](#) ([Best Paper Award](#)).
- [22] Pratik Gajane and Mykola Pechenizkiy. On Formalizing Fairness in Prediction with ML. In *the 5th Workshop on Fairness, Accountability, and Transparency in Machine Learning (FAT/ML)*, 2018 [↗](#).
- [23] Peter Auer, Pratik Gajane and Ronald Ortner. Adaptively Tracking the Best Bandit Arm with an Unknown Number of Distribution Changes. In *the 14th European Workshop on Reinforcement Learning (EWRL)*, 2018 [↗](#).
- [24] Pratik Gajane, Tanguy Urvoy and Emilie Kaufmann. Corrupt Bandits for Preserving Local Privacy. In *the proceedings of the 29th International Conference on Algorithmic Learning Theory (ALT)*, 2018 [↗](#).

- [25] Carolin Lawrence, Pratik Gajane and Stefan Riezler. Counterfactual Learning for Machine Translation: Degeneracies and Solutions. In *the workshop for Causal Inference and Machine Learning for Intelligent Decision Making at NeurIPS 2017* [↗](#).
- [26] Pratik Gajane, Tanguy Urvoy and Emilie Kaufmann. Corrupt bandits. In *the 13th European Workshop on Reinforcement Learning (EWRL)*, 2016 [↗](#).
- [27] Pratik Gajane, Tanguy Urvoy and Fabrice Clerot. A Relative Exponential Weighing Algorithm for Adversarial Utility-based Dueling Bandits. In *the proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015 [↗](#).
- [28] Pratik Gajane and Tanguy Urvoy. Utility-based Dueling Bandits as a Partial Monitoring Game. In *the 12th European Workshop on Reinforcement Learning (EWRL)*, 2015 [↗](#).

## PROJECTS

|                     |                                                                                                        |                              |
|---------------------|--------------------------------------------------------------------------------------------------------|------------------------------|
| SEP 2024 - PRESENT  | French National Research Agency (ANR)<br>For a project on responsible machine learning for healthcare. | Individual Award of €200,000 |
| FEB 2021 - MAY 2024 | Dutch Research Council (NWO) Project TOP TEPAIV <a href="#">↗</a>                                      | Co-leader of a Workstream    |
| FEB 2018 - JAN 2021 | CHIST-ERA Project DELTA <a href="#">↗</a>                                                              | Co-leader of a Workstream    |

## AWARDS AND RECOGNITION

- Runner-up Frontier Prize at the 21st Symposium on Intelligent Data Analysis (IDA), 2023.
- Travel grant of 3000 CAD awarded by the Alberta Machine Intelligence Institute to attend Upper Bound 2023, Edmonton, Canada.
- Best Paper Award at Lifelong Learning: A Reinforcement Learning Approach Workshop at FAIM 2018.
- Nominated for the Best Master's Thesis in Computer Science at the Indian Institute of Technology Madras, 2014.
- Top 0.25 percentile in the Graduate Aptitude Test in Engineering (Computer Science) India 2012.
- Won the 3<sup>rd</sup> prize in a national level robotics competition – Versatelia, 2008.

## TEACHING

|            |                                                             |     |               |                               |
|------------|-------------------------------------------------------------|-----|---------------|-------------------------------|
| 2022-23 Q2 | Data Mining (Lecturer)                                      | BSc | 140 students  | Eindhoven Univ. of Technology |
| 2022-23 Q1 | Reinforcement Learning (Lecturer)                           | MSc | 35 students   | Eindhoven Univ. of Technology |
| 2022-23 Q1 | Embodying Intelligent Behavior in Social Context (Lecturer) | MSc | 41 students   | Eindhoven Univ. of Technology |
| 2021-22 Q4 | Data Intelligence (Project Supervision)                     | MSc | 50 students   | Eindhoven Univ. of Technology |
| 2013-14 S2 | Data Mining (TA)                                            | BSc | ~ 20 students | IIT Madras                    |
| 2013-14 S1 | Introduction to Machine Learning (TA)                       | BSc | ~ 60 students | IIT Madras                    |
| 2012-13 S2 | Computational Engineering (TA)                              | BSc | ~ 50 students | IIT Madras                    |
| 2012-13 S1 | Introduction to Research (TA)                               | BSc | ~100 students | IIT Madras                    |

See some of my student evaluations in my UTQ/BKO dossier [↗](#).

A short snippet from my introductory lecture on reinforcement learning can be found at this link [↗](#).

## PEDAGOGICAL CERTIFICATIONS

|      |                                                  |                               |
|------|--------------------------------------------------|-------------------------------|
| 2023 | Assessment <a href="#">↗</a>                     | Eindhoven Univ. of Technology |
| 2023 | Evaluation of Courses <a href="#">↗</a>          | Eindhoven Univ. of Technology |
| 2022 | Teaching Skills <a href="#">↗</a>                | Eindhoven Univ. of Technology |
| 2022 | Designing Courses and Projects <a href="#">↗</a> | Eindhoven Univ. of Technology |
| 2022 | Facilitating Learning <a href="#">↗</a>          | Eindhoven Univ. of Technology |
| 2021 | Supervision of PhD Students                      | Eindhoven Univ. of Technology |

## SUPERVISION AND THESIS EVALUATION

Co-supervisor for following students:

### PhD

|              |                |                                                             |
|--------------|----------------|-------------------------------------------------------------|
| 2021-Present | Danil Provodin | Constrained Sequential Learning (in collaboration with KPN) |
|--------------|----------------|-------------------------------------------------------------|

### MSc

|           |                  |                                                                                 |
|-----------|------------------|---------------------------------------------------------------------------------|
| 2022-2023 | Joost v.d. Haar  | Supply Chain Management using ML (in collaboration with ASML) <a href="#">↗</a> |
| 2022-2023 | Jiong Li         | Exploration in Reinforcement Learning with Sparse Rewards <a href="#">↗</a>     |
| 2022-2023 | Wouter v. d. Wee | Curiosity-driven Fairness in Reinforcement Learning <a href="#">↗</a>           |

Member of the thesis evaluation committee for following students:

|     |                      |                                                                                     |
|-----|----------------------|-------------------------------------------------------------------------------------|
| MSc | Daan Roordink        | On Multivariate Distributional Regression Techniques and Evaluation Metrics         |
| MSc | Youri Vis            | Effective Sampling in Intrinsically Motivated Reinforcement Learning                |
| BSc | Adomas Repšys        | Explanation of the Inconsistencies in the Performance of Bias Mitigation Techniques |
| BSc | Shadiah R. Lacouture | Justifying the Use of Fairness Pre-Processing Algorithms                            |
| BSc | Shahrukh Tufail      | Auditing Recommender Models for Bias without Access to Sensitive Data               |

## PROFESSIONAL ACTIVITIES

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|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reviewer</b>          | JMLR, ICML, NeurIPS ( <a href="#">Top Reviewer</a>  ) , AAAI, ICLR, ACM Conference on Fairness, Accountability, and Transparency |
| <b>Program Committee</b> | UAI ( <a href="#">Top Program Committee Member 2022</a>  ) , ALT, IJCAI, ECML/PKDD, European Workshop on Reinforcement Learning  |

## LEADERSHIP AND OUTREACH ROLES

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|              |                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------|
| 2022-PRESENT | Academic Coach for the Honors Academy at Eindhoven University of Technology                                |
| 2021-PRESENT | Volunteer for the Diversity and Inclusion Task Force at Eindhoven University of Technology                 |
| 2016-2017    | Mentor to junior PhD students                                                                              |
| 2013-2014    | Volunteer tutor for high school students from disadvantaged households                                     |
| 2013-2014    | Student Head Coordinator for the Wellness and Outreach Initiative at Indian Institute of Technology Madras |
| 2012-2014    | Class Representative of MSc Computer Science class of 2014 at Indian Institute of Technology Madras        |

## SELECTED INVITED TALKS AND PRESENTATIONS

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|               |                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUG 05, 2024  | Tutorial on fair reinforcement learning at the 33rd International Joint Conference on Artificial Intelligence (IJCAI) 2024 at Jeju, South Korea  |
| MAY 21, 2024  | Invited virtual talk at the Conference on Advancing Behavioral Science through AI and Digital Health at the University Michigan, USA            |
| AUG 23, 2023  | Presentation at the 32nd International Joint Conference on Artificial Intelligence (IJCAI) 2023 at Macau, China                                 |
| APR 13, 2023  | Presentation at the Symposium on Intelligent Data Analysis at Louvain-la-neuve, Belgium                                                       |
| SEPT 4, 2019  | Invited talk at DeepMind, Google at London, UK                                                                                                  |
| DEC 28, 2018  | Invited talk at IIT Madras, Department of Computer Science at Chennai, India                                                                  |
| JULY 15, 2018 | Presentation at the 5th Workshop on Fairness, Accountability, and Transparency in Machine Learning at Stockholm, Sweden                         |
| JULY 14, 2018 | Presentation at the 2nd workshop for Lifelong Learning: A Reinforcement Learning Approach at Stockholm, Sweden                                  |
| APR 7, 2018   | Presentation at the 29th International Conference on Algorithmic Learning Theory at Lanzarote, Spain                                            |
| NOV 22, 2017  | Invited talk at Montanuniversität Leoben at Leoben, Austria                                                                                   |
| JUNE 7, 2017  | Invited talk at Heidelberg University, Statistical Natural Language Processing Colloquium at Heidelberg, Germany                                |
| JULY 07, 2015 | Presentation at the 32nd International Conference on Machine Learning (ICML) at Lille, France                                                 |

## SKILLS AND EXPERTISE

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|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>         | Python, C++, Java, R, MATLAB.                                                                                                                 |
| <b>ML Tools</b>            | TensorFlow, PyTorch, Jax, Scikit-Learn, XGBoost, Hugging Face Transformers, MLflow, Keras, Microsoft Azure ML, DevOps, MLOps.                 |
| <b>Platforms and Tools</b> | Unix/Linux, Docker, Kubernetes, Git, AWS, Apache Spark, SSH, Hadoop, Jupyter Notebook.                                                        |
| <b>Model Optimization</b>  | Experience in transfer learning, model compression, quantization, and optimization techniques for efficient deployment of large-scale models. |

## INDUSTRY EXPERIENCE

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|-----------|----------------------------------------------------------------------------|
| 2009-2011 | Infosys Information technology consulting company, <i>Systems Engineer</i> |
|-----------|----------------------------------------------------------------------------|